

# Interpolation Beats Finite-Size Scaling for Domain Extension of Learned Transfer-Operator Spectra: A Two-System Study and a Convergence Diagnostic

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## Abstract

Can one predict the long-time statistics of an extensively chaotic field on a large domain by learning a transfer operator only on small domains and extrapolating its spectrum in system size? We built the finite-size-scaling (FSS) machinery to do exactly this — estimate leading Ruelle–Pollicott / Koopman resonances and spectral densities by delay-embedded EDMD, fit a per-wavenumber  $1/L$  flow on small domains, extrapolate — and the honest verdict, across two systems chosen to bracket the finite-size convergence rate, is that it does *not* beat the simplest baseline: interpolating the spectrum of the largest affordable box in wavenumber, with no size flow at all. For Kuramoto–Sivashinsky (KS) the flow works in absolute terms —  $\{22, 44, 66, 88\} \rightarrow L = 1408$  ( $64\times$  the base volume, zero large-domain data) predicts decay rates to 9.8%, spectral density to 12.5%, correlations to 9.6% — but is *redundant*: KS spectra converge by  $L = 88$  (drift 4.3%), so interpolation does better (3.3%). We then ran the Nikolaevskiy equation, whose marginal  $k=0$  conservation mode makes the *spectrum* converge slowly (drift 35.0% at  $L = 88$ ): here interpolation is genuinely degraded (16.3% on decay rates at  $L = 704$ ) — yet the  $1/L$  flow is *worse still* (21.4%), because the spectral density stays extensive and the flow overshoots the few non-monotone, non-converged low- $k$  decay rates. The FSS flow has no advantage window over interpolation in either regime. The operative diagnostic is not the correlation length (short,  $\approx 4.7$  and 6, for both) but the *small-domain spectral drift*, which a practitioner can measure to predict interpolation’s accuracy. A deep translation-equivariant Koopman autoencoder reproduces the invariant density (8.2%) but not per-mode resonances (328.3% error), so EDMD is our instrument. Every number is machine-generated and regenerates byte-for-byte.

## 1 Introduction

Extensively chaotic spatiotemporal systems have a thermodynamic limit: as the domain grows, intensive statistics (Lyapunov-exponent density, correlation lengths, spectral densities) converge, with finite-size corrections that vanish as the box grows [Cross and Hohenberg, 1993, Egolf and Greenside, 1994, Manneville, 1985]. Transfer-operator (Ruelle–Pollicott, RP) resonances [Ruelle, 1986, Pollicott, 1985] encode the statistics that survive to long times in a chaotic system — correlation-decay rates, power spectra, mixing — and are, unlike trajectories, legitimately predictable past the Lyapunov horizon. This paper asks a scaling question about them: *does the size-dependence of a learned operator’s spectrum converge smoothly enough that one can fit it on small domains and extrapolate the statistics of a much larger one, without ever simulating or training at the large size?*

This connects three threads of our own program: transfer-operator statistics as the right target for surrogate models of chaos [Howe, 2026b]; long-horizon statistical (not trajectory) fidelity [Howe, 2026a]; and scaling, rather than brute-force training, as the route to large systems. Two narrowings keep the claim honest. (N1) What is predicted at long horizon is *statistics and slow modes*, *not trajectories*:  $e^{\lambda t}$  spectral continuation cannot beat a positive Lyapunov exponent, but RP

resonances do give correlation functions, power spectra, and decay rates [Chekroun et al., 2020]. (N2) Koopman/RP spectral *estimation and forecasting* is an occupied field [Chekroun et al., 2020, Arbabi and Mezić, 2017, Das and Giannakis, 2019, Williams et al., 2015]; we treat it as substrate. The only mechanism we test as novel is *domain-size extrapolation* of a learned operator’s spectrum via a learned finite-size-scaling flow.

**Contributions.** (i) A characterization (§4): the finite-size dependence of EDMD-estimated KS RP/Koopman spectra is smooth and  $1/L$ -fittable (*GATE S passes*); to our knowledge the first such measurement, and the foundation any FSS flow would stand on. (ii) A working FSS flow (§6) that does extrapolate the spectrum to  $64\times$  the base volume at single-digit-percent error from zero large-domain data. (iii) **The honest verdict:** across two systems chosen to bracket the finite-size convergence rate, interpolating the largest affordable box beats the flow — for KS because the spectrum is converged and the flow is redundant (§7), and for the Nikolaevskiy equation, whose marginal mode makes the spectrum converge slowly, because the flow overshoots the non-monotone drift while the spectral density stays extensive (§9). The flow has no advantage window; we say so and give the mechanism. (iv) **A diagnostic:** the small-domain spectral drift (not the correlation length, which is short for both systems) is measurable from the training data and predicts interpolation’s accuracy, so a practitioner can tell whether extrapolation is trustworthy before paying for a large run. (v) An honest negative for deep operators (§5): a translation-equivariant Koopman autoencoder matches the invariant measure but not per-mode resonances, so we scale the EDMD operator.

## 2 Related work

**RP resonances of stochastic/observed systems.** Chekroun et al. [2020], Tantet et al. [2020a,b] develop reduced-state-space RP resonances (with an El-Niño application); our substrate is this machinery, estimated from data. **Koopman spectral estimation and forecasting.** EDMD [Williams et al., 2015], delay-coordinate / Hankel Koopman [Arbabi and Mezić, 2017, Das and Giannakis, 2019, Mezić, 2005, Takens, 1981], DMD for analytic maps [Slipantschuk et al., 2020], and Koopman autoencoders [Lusch et al., 2018, Otto and Rowley, 2019, Azencot et al., 2020] are the fixed-domain tools we build on. **Domain extension for neural operators** is the named adjacent problem: de Villeroché et al. [2026] achieve zero-shot generalization of transformer neural operators to larger domains via attention locality, and Huang et al. [2025] via domain-decomposition operator learning; both are *locality* routes. Ours is a *spectral / renormalization* route, and our finding is that a plain wavenumber-interpolation null already matches or beats the spectral flow in both convergence regimes we tested — so we cite-compare rather than reimplement. **RG  $\times$  ML:** neural-network renormalization group [Li and Wang, 2018] and analyses of when learning dynamics renormalize [Zhang, 2025] share the RG framing but target sampling / training dynamics, not operator-spectrum domain extension. Our inspiration is Wilsonian momentum-shell scaling; empirically the KS data support only a smooth-in- $1/L$  fit, and we do not oversell the RG language. Dynamic-RG analyses of the *noisy* KS equation place its long-wavelength field in the KPZ universality class [Ueno et al., 2005]; that is a different object (field scaling exponents), not the finite-size dependence of the transfer-operator spectrum we measure.

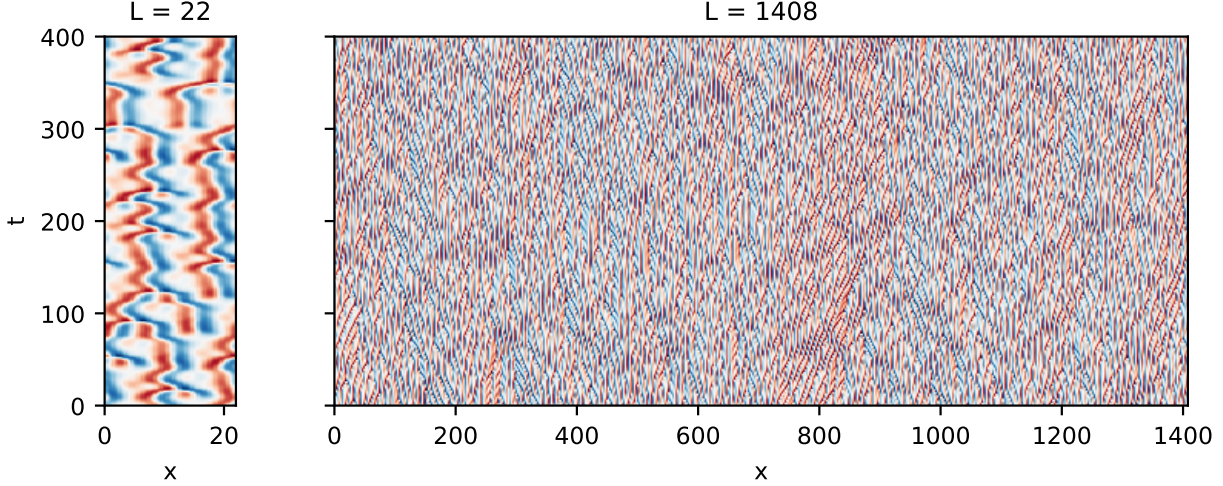


Figure 1: KS space-time fields on the base domain ( $L = 22$ ) and the target ( $L = 1408 = 64\times$ ). The intensive statistics — correlation length, spectral density per unit length — have a thermodynamic limit; this paper asks whether the finite-size dependence of the learned transfer-operator spectrum can be fit on small domains and extrapolated to the large one.

### 3 Setup

We integrate KS,  $u_t = -uu_x - u_{xx} - u_{xxx}$ , periodic on  $[0, L)$ , with a pseudospectral ETDRK4 scheme [Kassam and Trefethen, 2005] (2/3 dealiasing), vendored and regression-pinned against a prior validated implementation [Howe, 2026b]. The grid resolution  $\Delta x = 22/64$  is fixed for all sizes,  $dt = 0.25$ , samples every 0.5 time units after a 500-unit transient; 3 seeds per size.

**Per-sector spectra.** Translation invariance block-diagonalizes the dynamics over Fourier wavenumbers  $k = 2\pi m/L$ . For each retained sector ( $0 < k \leq 3$ ) we form the complex mode series  $\hat{u}_m(t)/N$  and estimate its leading resonance by Hankel-EDMD: an 8-delay dictionary at stride  $s$  samples, one-step map advancing by  $s$ , Grams accumulated in a single streaming pass over  $\{1, 8, 64, 512\}$ ; the stride whose window is closest to  $3\tau_e(k)$  (with an anti-aliasing guard) is selected. The leading resonance  $\lambda(k) = \log \mu/(s \Delta t_s)$  gives decay rate  $\gamma(k) = -\text{Re } \lambda$  and frequency  $\omega(k) = |\text{Im } \lambda|$ . We also accumulate a Welch power spectrum; the spectral density is  $\hat{S}(k) = P_m/(2\pi/L)$ , the object with a thermodynamic limit. Over all sizes the EDMD-implied autocovariance matches the measured one with median  $R^2$  between 0.94 and 0.99.

### 4 E0: finite-size scaling of KS spectra exists (GATE S)

On the common grid  $k = 2\pi m/22$  (present at every training size) we fit, per wavenumber,  $y(L) = y_\infty + c(22/L)^p$  with  $p \in \{1, 2\}$  chosen by AICc, seed-averaged and SE-weighted. **GATE S** (pre-registered): a smooth or already-converged size-dependence exists for  $\geq 5/7$  wavenumbers (passed:  $6/7$  for  $\gamma$ ,  $7/7$  for  $\hat{S}$ ), and fitting on  $\{22, 44, 66, 88\}$  predicts held-out  $L = 176$  to  $\leq 10\%$  (achieved:  $8.2\%$  for  $\gamma$ ,  $2.1\%$  for  $\hat{S}$ ). **GATE S passes.** Finite-size corrections are large at small  $L$  — at  $k \approx 0.57$  the decay rate grows from  $0.015$  ( $L = 22$ ) to  $0.087$  ( $L = 88$ ), a factor  $5.7$  — and mostly saturate by  $L = 88$ ; AICc prefers the  $1/L^2$  form for  $4/7$  of the density points. This smooth, measurable size-dependence is the paper’s foundation (Fig. 2).

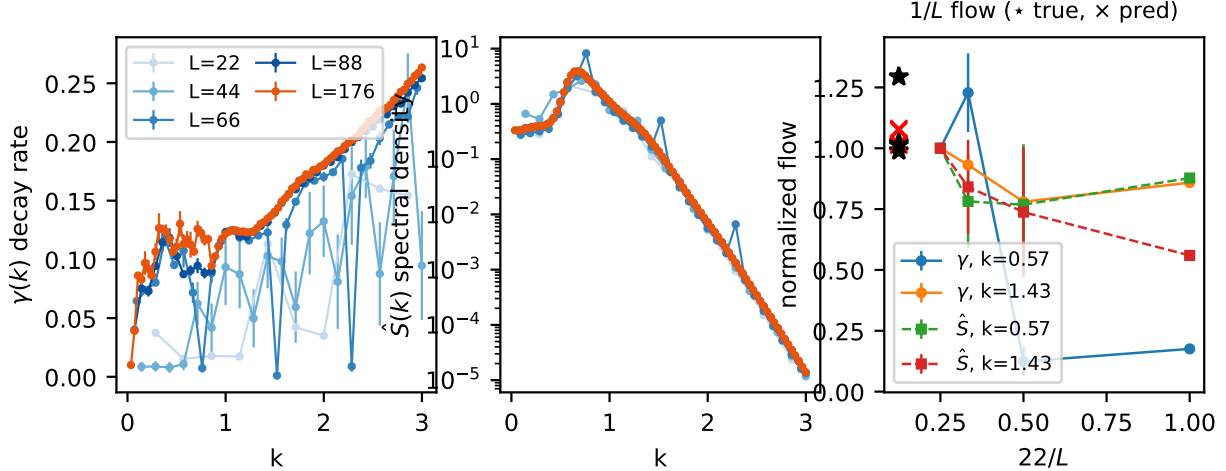


Figure 2: E0 characterization. Left: decay rate  $\gamma(k)$ ; middle: spectral density  $\hat{S}(k)$ ; both vs. wavenumber for  $L \in \{22, 44, 66, 88, 176\}$ . Right: the per-wavenumber  $1/L$  flow at two common wavenumbers, normalized;  $\star$  is the held-out  $L = 176$  truth and  $\times$  the flow prediction. GATE S passes.

## 5 E1: a deep Koopman operator matches the measure, not the resonances

We train a translation-equivariant convolutional Koopman autoencoder (circular convolutions, a linear circular-convolution propagator, 16 latent channels) at  $L = 22$ , 11.2 min/model. It reproduces the *invariant spectral density* well — generated fields match the measured density to 8.2% (median) and the two-point correlation to 12.6% — but its per-sector *leading resonances* do not match EDMD: even under generate-and-reestimate (drive the learned stochastic operator, run the identical estimator used on data), the decay rates are off by 86.5% and the oscillation frequencies by 328.3%. The variance-dominated reconstruction loss pins the measure without pinning each mode’s resonant phase. Per our pre-registered kill condition K3, we therefore build the FSS flow on the data-driven Koopman (EDMD) spectrum — itself a learned transfer operator — and report the deep operator as this measure-fidelity/resonance-infidelity result.

## 6 E2: the finite-size-scaling flow works

The FSS flow generalizes the GATE-S recipe to arbitrary target wavenumbers: for each target  $k$ , gather the value at the nearest mode of each training size and fit the seed-averaged  $1/L$  (or  $1/L^2$ , AICc) model, extrapolating to the target  $L$ . We report the headline over the *full-support band*  $k \in [2\pi/22, 2.2]$  (wavenumbers present at all four training sizes); the partial-support band  $[2\pi/88, 2\pi/22]$  and the genuinely new long-wave modes ( $k < 2\pi/88$ ) are reported separately.

Fitting on  $\{22, 44, 66, 88\}$  and extrapolating with *zero* large-domain data, the flow predicts the  $L = 1408$  spectrum (672 sectors,  $64\times$  the base volume): decay rates to 9.8%, spectral density to 12.5%, band-limited two-point correlation to 9.6%, correlation times to 6.1%, and a slow-mode-subspace Jaccard overlap of 0.38 (Fig. 3). The new long-wave band is revealing: its *spectral density* extrapolates almost perfectly (4.1% over 15 modes, since  $\hat{S}$  is flat at low  $k$ ) but its *decay rates* do not (394.8% error) — these hydrodynamic modes are far slower than anything in the training set, so the dynamics, unlike the statics, cannot be extrapolated in wavenumber. Taken alone this is a

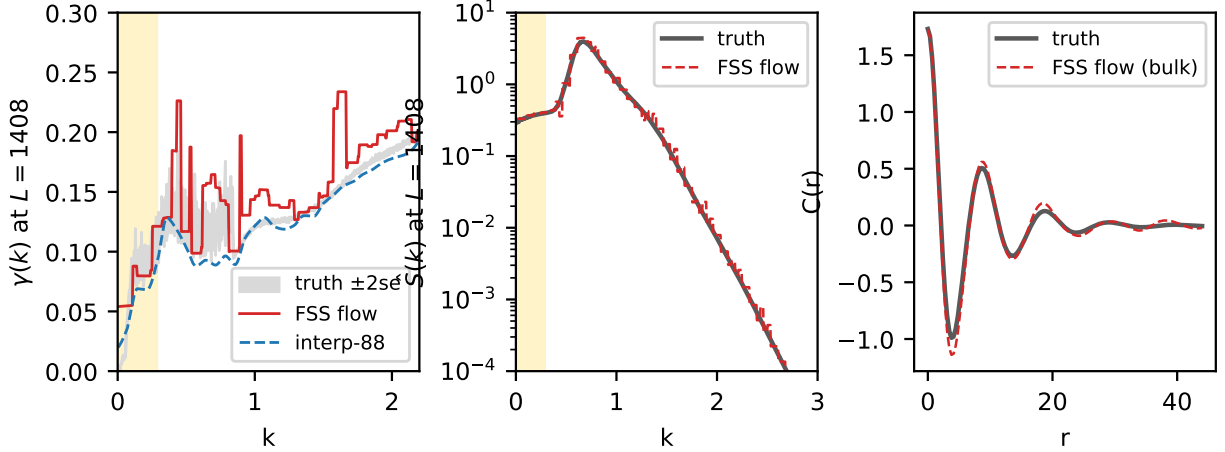


Figure 3: E2 domain extension to  $L = 1408$ . Decay rate  $\gamma(k)$ , spectral density  $\hat{S}(k)$ , and two-point correlation  $C(r)$ : truth (grey), FSS flow (red), and the no-flow null interp-88 (blue, left). Both are fit only on  $L \leq 88$ ; the shaded band is the sub- $2\pi/22$  region (partial/new-mode support) reported separately.

positive domain-extension result; §7 shows why it is nonetheless unnecessary.

## 7 E3: KS — when interpolation suffices (K2 fires)

The flow works, but does it *beat* the obvious no-flow null? For KS, no — and understanding why gives half the criterion. We compare the flow against the strongest no-flow null: **interp-88**, which interpolates the measured  $L = 88$  spectrum in wavenumber to the target grid with *no size flow at all* (the spectral analog of evaluating a small-domain operator on a larger domain, the locality/zero-shot route [de Villeroché et al., 2026]). Table 1 is decisive: interp-88 predicts the  $L = 1408$  decay rates to 3.3%, density to 0.4%, and correlations to 1.8% — *better* than the flow on all 5 headline metrics. Our pre-registered kill condition **K2 fires** (the flow wins 0/5): the FSS mechanism adds nothing over translation-invariant interpolation for KS.

The mechanism is fast convergence (§8, Table 2): the KS bulk spectrum is within 4.3% (decay rates) and 0.5% ( $\hat{S}$ ) of the  $L = 1408$  limit *already at*  $L = 88$ , so using the largest affordable box directly beats extrapolating a  $1/L$  flow, which over-corrects past the already-converged largest size. The weaker strict-tiling null (periodically tile the  $L = 22$  field) captures band-integrated power to 52.5% but fails on the two-point correlation (137.2%) through the spurious  $L = 22$  periodicity it imposes — a reminder that tiling, unlike smooth  $k$ -interpolation, injects artificial long-range order. A flow fit on an aggressive base  $\{22, 33, 44\}$  is worse still (87.3% on decay rates; its density extrapolation is numerically unstable), and even there interp-44 wins. With *some* large-domain data, direct EDMD at the target reaches 2.3% (decay rates) from only 10,000 time units — and the full oracle costs  $8.0\times$  our small-domain compute, while interp-88 costs the same as the flow.

## 8 E4: boundaries — where each route breaks

**Convergence: the diagnostic that matters.** Table 2 reports the median distance of each size’s common-grid spectrum from the  $L = 1408$  limit — the *spectral finite-size drift*. The KS decay rates are within 56.8% by  $L = 44$  and 4.3% by  $L = 88$ ; this rapid convergence is exactly why interp-88

Table 1: Domain extension to  $L = 1408$  ( $64\times$  the base volume), headline band  $k \in [2\pi/22, 2.2]$ . Columns: median relative error of decay rate  $\gamma$  (%), median  $|\log_{10}|$  of the density ratio, band-limited  $C(r)$  relative  $L_2$  (%), correlation-time  $\tau_e$  error (%), slow-mode Jaccard overlap. The FSS flow works, but the no-flow null *interp-88* does better on every metric: **K2 fires**. All values are the seed-mean (GATE-S) point estimate. The strict-tiling  $C(r)$  is reconstructed from the tiled *full-field* power (tiling is defined on the field, not the retained band), whereas the flow and interpolation  $C(r)$  are band-limited to the headline band on both sides; the tiling entry is therefore not column-comparable and is shown only to expose its spurious-periodicity failure.

| method                   | $\gamma$ | $\hat{S}  \log_{10} $ | $C(r)$ | $\tau_e$ | slow |
|--------------------------|----------|-----------------------|--------|----------|------|
| FSS flow (ours)          | 9.8      | 0.051                 | 9.6    | 6.1      | 0.38 |
| FSS flow, base $\leq 44$ | 87.3     | 1.347                 | 1559.7 | 32.5     | 0.00 |
| interp-88 (null)         | 3.3      | 0.002                 | 1.8    | 4.7      | 0.46 |
| interp-44 (null)         | 51.2     | 0.111                 | 31.5   | 30.8     | 0.38 |
| interp-22 (null)         | 80.3     | 0.136                 | 38.5   | 146.9    | 0.36 |
| EDMD@ $L$ , $T=2000$     | 45.2     | 0.008                 | 0.5    | 63.0     | 0.29 |
| EDMD@ $L$ , $T=10000$    | 2.3      | 0.004                 | 0.4    | 1.7      | 0.39 |
| strict tiling (null)     | –        | –                     | 137.2  | –        | –    |

Table 2: Convergence of the KS bulk spectrum to the  $L = 1408$  limit: median relative distance (%) on the common grid  $k = 2\pi m/22$ . The decay rates are within a few percent of the limit already by  $L = 88$ ; this fast convergence is why interpolating the largest small box (interp-88) beats the finite-size flow.

| $L$ | $\gamma$ to limit (%) | $\hat{S}$ to limit (%) |
|-----|-----------------------|------------------------|
| 22  | 81.1                  | 22.0                   |
| 44  | 56.8                  | 23.2                   |
| 66  | 9.2                   | 19.9                   |
| 88  | 4.3                   | 0.5                    |
| 176 | 2.4                   | 0.3                    |
| 352 | 0.8                   | 0.2                    |
| 704 | 0.4                   | 0.2                    |

suffices and the flow is redundant. The drift is measurable from the training sizes alone, so it, rather than any post-hoc knowledge, tells the practitioner interpolation is safe here. One might expect the real-space correlation length to be the diagnostic instead; it is not. The KS integral correlation length is  $\xi \approx 4.7$  (and  $C(r)$  is negligible beyond  $r \approx 40$ ), but so is Nikolaevskiy’s (6, §9) — yet the two converge at completely different rates. What separates them is a marginal mode, not a long correlation length; the drift captures its effect directly.

**Extrapolation distance.** Across the ladder  $L \in \{176, \dots, 2816\}$  (Table 3) both the flow and interp-88 stay flat in error (flow 9.8%→9.6% over  $1408 \rightarrow 2816$ ; interp-88 3.3%→3.8%), consistent with a converged bulk spectrum: distance in  $L$  is not the limiting factor once past  $L \approx 88$ .

**Boundary conditions.** Under odd-parity (Dirichlet-type  $u = u_{xx} = 0$ ) boundaries at  $L = 176$ , the bulk two-point correlation heals to the periodic prediction within a wall layer of 6.5 units; the periodic FSS-flow prediction reproduces the odd-parity *bulk* correlation to 7.6% and the bulk-point temporal autocorrelation to 12.5%, i.e. the size-scaled periodic spectrum transfers to the bulk of a non-periodic domain.

**Scale content.** Splitting  $L = 1408$  into an energy-containing band ( $k \leq 1.5$ ) and a microscale band ( $k > 1.5$ ), the flow’s decay-rate error is 11.7% in the energy band and 6.3% in the microscale band.

**New long-wave modes.** The genuinely new modes ( $k < 2\pi/88$ , absent at every training size)

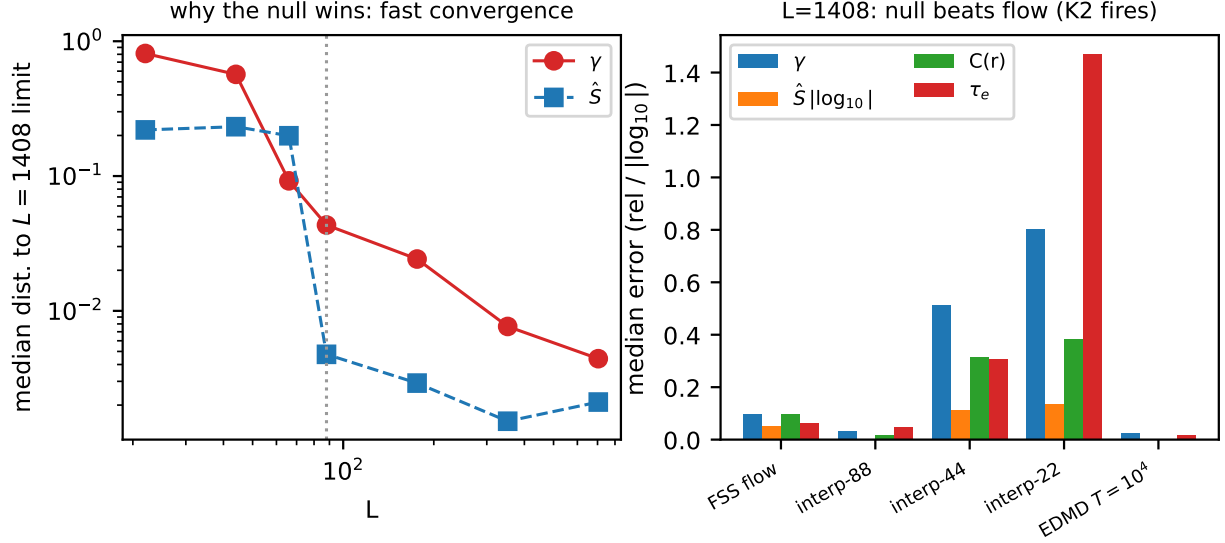


Figure 4: Left: the mechanism — the KS bulk spectrum’s median distance to the  $L = 1408$  limit collapses to a few percent by  $L = 88$  (dotted line), so the largest small box is already near-converged. Right: at  $L = 1408$ , the no-flow null interp-88 beats the FSS flow on every headline metric — K2 fires.

Table 3: Extrapolation distance: decay-rate median relative error (%) vs. target  $L$  (both routes fit only on  $L \leq 88$ ). Error is flat past  $L \approx 88$  for both, consistent with an already-converged bulk spectrum; interp-88 leads throughout.

| method    | 176 | 352 | 704 | 1408 | 2816 |
|-----------|-----|-----|-----|------|------|
| FSS flow  | 7.6 | 9.7 | 9.6 | 9.8  | 9.6  |
| interp-88 | 1.9 | 2.9 | 3.2 | 3.3  | 3.8  |

separate statics from dynamics cleanly (Fig. 6): their spectral density extrapolates in wavenumber to 4.1% (the density is flat at low  $k$ ), but their decay rates, being far slower than anything trained, are not recovered (394.8%). No method — flow or null — predicts the new modes’ *dynamics*; this is the honest ceiling of size-and-wavenumber extrapolation.

## 9 E5: Nikolaevskiy — a slow-converging spectrum, and the flow still loses

KS converges so fast that interpolation is near-exact; does the flow earn its keep when the spectrum converges *slowly*? We test the hardest honest case: the Nikolaevskiy equation,  $u_t = -u u_x + \mathcal{F}^{-1}[\sigma(k) \hat{u}]$  with linear growth rate  $\sigma(k) = k^2[r - (1 - k^2)^2]$ . The prefactor  $k^2$  makes  $k=0$  *exactly marginal* — a conservation-law/Goldstone mode — giving soft-mode turbulence [Nikolaevskii, 1989, Tribelsky and Tsuboi, 1996, Matthews and Cox, 2000] in which the marginal mode renormalizes the low- $k$  decay rates with large, slow finite-size corrections. We run it at  $r = 0.10$  with the identical integrator family, estimators, and flow (ladder {44, 66, 88};  $L = 22$  is a turbulence-onset outlier at this  $r$  and is excluded). The effect is real and large: the decay-rate drift is still 35.0% at  $L = 88$  (Fig. 7, left), roughly  $8\times$  KS’s 4.3%, so **interpolation is genuinely degraded** — interp-88 misses the  $L = 704$  decay rates by 16.3% (versus 3.3% for KS). Notably the real-space correlation length

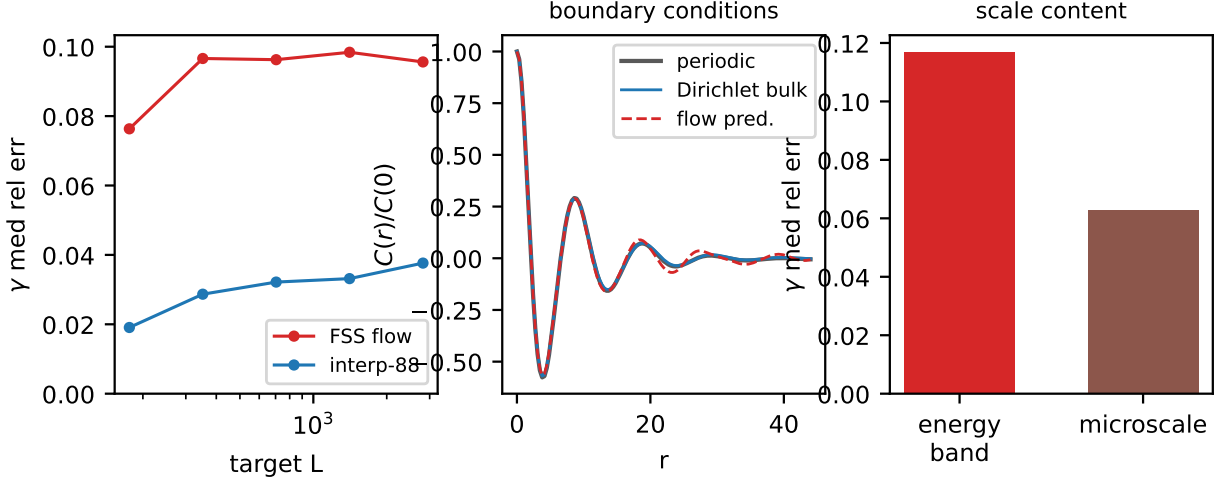


Figure 5: E4. Left: decay-rate error vs. target  $L$  (flat past  $L \approx 88$  for both routes). Middle: odd-parity (Dirichlet-type) bulk correlation matches the periodic FSS-flow prediction outside a thin wall layer. Right: flow error split by scale band at  $L = 1408$ .

is *not* what changed: it is  $\xi \approx 6$ , essentially the same short value as KS. The slow convergence is a spectral effect of the marginal mode, not a long correlation length.

This is the regime the flow was built for — and it still loses. Extrapolating the flow from  $\{44, 66, 88\}$  to  $L = 704$  (336 sectors, zero large-domain data) gives decay rates to 21.4% — *worse* than the degraded interp-88 (16.3%) — and spectral density to 27.5% versus interp-88’s 1.1%. The flow beats the best no-flow null on 0/5 metrics. Two mechanisms defeat it, both visible in the data. First, the spectral density is *extensive* even here:  $\hat{S}(k)$  is converged by  $L = 44$ , so interpolation is near-exact for it (1.1%) while the flow only adds extrapolation noise (27.5%). Second, the finite-size drift that does exist — in the low- $k$  decay rates — is *non-monotone* across the training sizes (soft-mode-contaminated, and estimated from finite trajectories), so the  $1/L$  fit, extrapolated over an  $8\times$  span in system size, overshoots; interp-88, which does no extrapolation, is closer despite being biased. The flow needs a clean monotone approach to a limit, and the marginal-mode regime does not provide one. So across both systems — fast-converging (KS) and slow-converging (Nikolaevskiy) — interpolating the largest affordable box beats the  $1/L$  flow.

## 10 What we do not claim, and where the flow might yet help

We do not claim trajectory forecasting (N1): only statistics and slow modes. We do *not* claim the FSS flow beats interpolation — for neither KS nor Nikolaevskiy does it, and the honest reading is that a  $1/L$  finite-size-scaling flow has no advantage window over interpolating the largest affordable box for these 1-D operator spectra. We make no 2-D/3-D or turbulence claims (none were run) and no claim about universality classes. We do not claim the deep operator captures resonances (it does not). We do not claim the new long-wave modes’ *dynamics* are predicted — only their spectral density.

**Why the flow lost, and the one regime that could reverse it.** The flow is redundant when the spectrum converges fast (interpolation is then near-exact) and it overshoots when convergence is slow but *non-monotone* (as the marginal-mode regime makes it). The gap it would need is

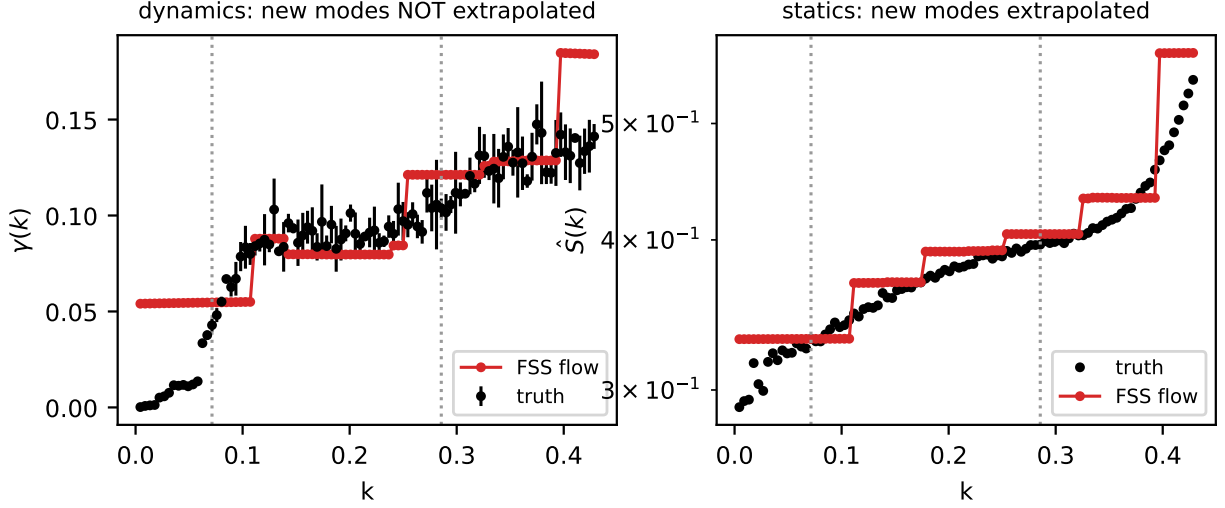


Figure 6: The new long-wave band at  $L = 1408$  ( $k < 2\pi/22$ , dotted lines mark  $2\pi/88$  and  $2\pi/22$ ). Left: decay rates of the genuinely new modes are not extrapolated. Right: their spectral density is, because  $\hat{S}$  is flat at low  $k$ . Statics extrapolate in wavenumber; dynamics do not.

a system whose spectrum converges *slowly but cleanly and monotonically* in  $1/L$  — extensive, so a thermodynamic limit exists to extrapolate to, yet far from it at the largest affordable box. Neither of our two systems is that: KS converges too fast, and Nikolaevskiy’s slow convergence is non-monotone and confined to a few soft modes while its density stays extensive. Whether such a clean-slow-convergence system exists (long-range coupling and size-dependent instabilities are the candidates we could not rule out) is the open question our negative result sharpens. Until one is exhibited, the practical guidance is concrete: measure the small-domain spectral drift; if it has saturated, interpolate the largest box; if it has not, no fixed-form extrapolation we tested is trustworthy — acquire large-domain data.

## 11 Reproducibility

Every number above is emitted by `gen_paper_numbers.py` from committed JSON and regenerates byte-for-byte (`verify_regen.py`); the KS and Nikolaevskiy integrators, the streaming estimators, and the scaling fits are unit-tested (spectra vs. analytic linear sectors; KS integrator regression against a prior validated code; Nikolaevskiy marginal-mode, mean-conservation and chaos checks; scaling round-trip). Both systems share one ETDRK4 core and one system-agnostic estimator/flow pipeline — only the linear symbol changes — which is what makes the KS-vs-Nikolaevskiy comparison a controlled one. All runs use a fixed grid, 3 seeds, and pre-registered gates (PLAN.md). Code and data: DOI to be assigned at submission. Recency re-sweep logged in PLAN.md.

## References

Hassan Arbabi and Igor Mezić. Ergodic theory, dynamic mode decomposition, and computation of spectral properties of the Koopman operator. *SIAM Journal on Applied Dynamical Systems*, 16(4):2096–2126, 2017.

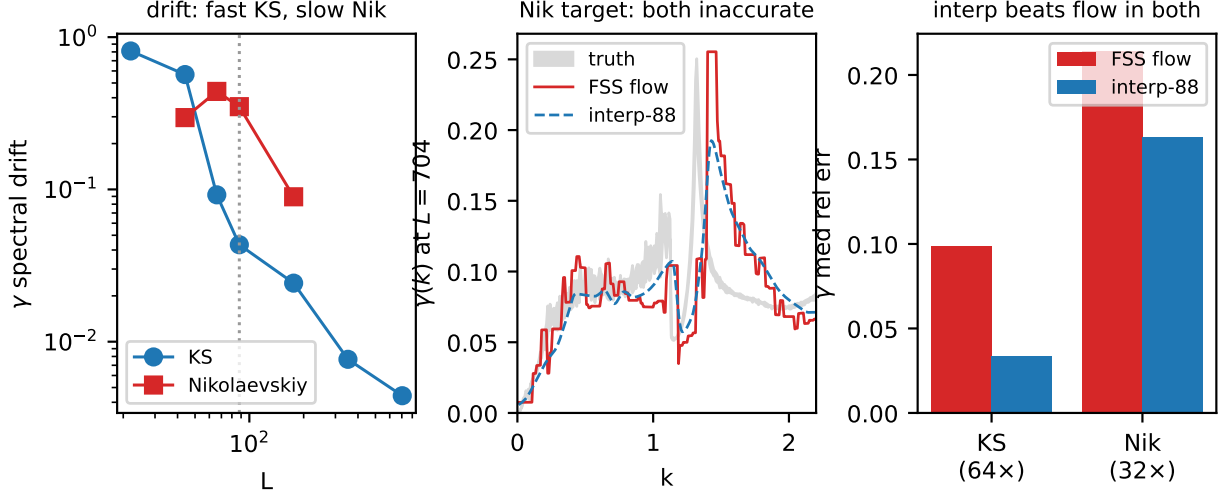


Figure 7: Two systems bracketing the convergence rate. Left: spectral drift (median distance of the bulk spectrum to the large- $L$  limit) vs.  $L$  — KS collapses by  $L \approx 88$  (dotted), Nikolaevskiy stays large. Middle: at the Nikolaevskiy target neither route is accurate for  $\gamma(k)$ ; the flow (red) overshoots the low- $k$  drift that interp-88 (blue) merely under-resolves. Right: decay-rate error of flow vs. interp-88 for both systems — interpolation wins in both, redundantly for KS and by overshoot-avoidance for Nikolaevskiy.

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